

DREW T. ERNST

Senior Systems Engineer | Data Analyst | Full-Stack Architect Quincy, IL | (217) 653-9445 |
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PROFESSIONAL PROFILE

Strategic Systems Engineer and Data Analyst with a track record of architecting high-stakes, production-grade applications. Expert in bridging the gap between complex data logic and intuitive user experiences, leveraging dual degrees in **Public Relations and Marketing** to deliver high-integrity solutions. Proven expertise in HIPAA-compliant clinical engines, **behavioral AI fintech**, and advanced SQL analytics. Specializes in finding the rhythm in messy data and building the technical pipelines to make it actionable for executive decision-making.

TECHNICAL TOOLKIT

- **Languages:** TypeScript (Strict Typing), JavaScript (ES6+), SQL (PostgreSQL, MySQL), Python (Pandas/NumPy), Java, C#.
 - **Frameworks & UI:** React 19, Next.js, Node.js, FastAPI, Vite, Tailwind CSS, Recharts, Framer Motion.
 - **Data Engineering:** Star Schema Design, ETL Pipelines, Window Functions (CTEs), RFM Segmentation, Data Normalization.
 - **Cloud & DevOps:** AWS (S3, EC2), Docker, CI/CD (GitHub Actions/GitLab), Nginx, Vercel, npm/pnpm.
 - **AI & Specialized:** Generative AI Orchestration (GPT-4o), Behavioral Mirroring, HIPAA Compliance, ACID Compliance.
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CORE ENGINEERING & ANALYTICS INITIATIVES

Principal Systems Architect • DTE Solutions LLC | 2025 – Present

Orchestrating technical infrastructure and BI strategy for high-performance ecosystems.

- **Business Intelligence Strategy:** Established a specialized analytical engine using **Next.js** and **Node.js** to transform raw operational metrics into actionable executive insights.
- **Architectural Scalability:** Refactored legacy codebases into modular, flattened structures, increasing developer velocity by 40% and reducing technical debt.
- **Full-Lifecycle Ops:** Managed production-grade deployments on **AWS (EC2/S3)** using **Docker** and **GitHub Actions**, maintaining 99.9% uptime for analytical pipelines.

Lead Data Analyst • E-Commerce SQL Analytics | 2026

Advanced SQL engineering and EDA repository for large-scale commerce telemetry.

- **Advanced Query Engineering:** Developed complex SQL models utilizing **Window Functions (NTILE, DENSE_RANK, LAG)** and CTEs for deep-dive customer segmentation and MoM growth tracking.

- **Cohort & LTV Analysis:** Engineered an automated cohort retention model to track **Lifetime Value (LTV)** and user "stickiness" across multi-stage purchase cycles.
- **Data Integrity Protocol:** Automated hygiene scripts to prune 2,000+ orphan records and normalize categorical inconsistencies, ensuring 100% downstream data accuracy.

Lead Architect // Data Engineer • Pulse-Ai (Behavioral Finance) | 2026

High-fidelity behavioral finance system bridging transactional data and psychological triggers.

- **Behavioral ETL Pipeline:** Engineered an asynchronous ETL process using **Python (Pandas)** to normalize bank logs into a **PostgreSQL Star Schema**.
- **Predictive Analytics:** Implemented impulse-trigger detection by correlating spending velocity with user stress indices (**Correlation vs. Causation**).
- **Secure Persistence:** Integrated **AWS S3** receipt ingestion using encrypted pre-signed URL protocols to ensure audit-ready security and user privacy.

Senior Software Engineer • Marion County DDSD (Missouri DMH) | 2026

PCSP Assistant Pro — HIPAA-compliant clinical documentation engine for state audits.

- **Compliance Engineering:** Developed a documentation workflow that reduced plan revision cycles by 40% while ensuring 100% adherence to Missouri DMH audit standards.
- **Logic Translation:** Built a client-side "Intelligent Phrasing Engine" in **Vanilla JavaScript**, translating complex clinical requirements into deterministic UI patterns.
- **Zero-Footprint Security:** Architected an in-memory processing system to satisfy stringent HIPAA security standards without compromising speed or fidelity.

Full-Stack Developer • NestLegacy | 2026

Digital Trust Layer and Lead Intelligence Engine for \$84T Wealth Transfer.

- **AI-First Intelligence:** Built a lead qualification engine using **GPT-4o** to perform **Categorical-to-Ordinal** mapping of prospective client behaviors.
- **Interactive Modeling:** Architected longevity and tax-exposure systems in **Next.js**, delivering high-precision mathematical value to financial fiduciaries.

EDUCATION & TRAINING

- **Advanced Engineering Mentorship:** David Bragg (Senior Software Engineer) | 2025 – 2026
 - *Intensive focus on enterprise architecture, OOD principles, and data orchestration.*
- **Frontend Simplified Bootcamp:** Production-Grade React & Software Architecture | 2025
- **B.S. Communication (Public Relations) // B.S. Marketing:** Quincy University & Culver-Stockton College
 - *GPA: 3.8 | S.I.F.E (Students in Free Enterprise) Strategic Lead.*